RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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USN										

Academic year 2024-2025 (Even Sem) CIE - 1 - 2025

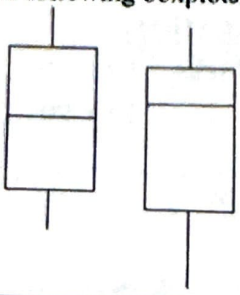
DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**01/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM241AT**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I**	
STATISTICS FOR DATA ANALYTICS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1	40th percentile in a given set of 140 data values occurs at position _____	(1)	L2	CO1
2	<table><tr><td>Class</td><td>21"-30"</td><td>31"-40"</td><td>41"-50"</td><td>51"-60"</td><td>61"-70"</td></tr><tr><td>No of occurrences</td><td>12</td><td>16</td><td>22</td><td>25</td><td>18</td></tr></table> The median in the above data lies in the class _____	(1)	L2	CO1
3	According to Bayes' rule, P(A|B) = ___, for two events in a sample space S.	(1)	L1	CO1
4	X = the no. coming up when a 6-faced die is thrown. Y = no. of heads when a coin is tossed. The possible values of X - Y are _____.	(1)	L3	CO2
5	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A′∩ B) = ___.	(1)	L3	CO2
6	P(X <= 3| Y=0) is ___, for X and Y as in Question no. 1.4	(1)	L3	CO2
7	A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective? _____	(2)	L3	CO1
8	Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: <table><tr><th rowspan="2">Type of unsaturation</th><th colspan="2">Type of oil</th></tr><tr><th>Canola</th><th>Corn</th></tr><tr><td>Mono</td><td>7</td><td>13</td></tr><tr><td>Poly</td><td>93</td><td>77</td></tr></table> If a bottle of oil is selected at random, what is the probability that it belongs to the polyunsaturated category?	(2)	L3	CO1
	Part –B			
1	Fourteen measurements of furnace temperatures (in ^{0}F) are recorded on successive batches in a semiconductor manufacturing process as follows: 953, 950, 948, 955, 951, 949, 957, 954, 955, 949, 952, 931, 953, 955. a) Calculate sample mean, sample variance, and standard deviation. b) Find the mode and median. How much could the largest temperature measurement be increased without changing the median value? c) Construct a box plot of the data, showing clearly the outlier bounds. d) Comment on the symmetry of the data.	(10)	L3	CO1
2 a	Describe any similarities or differences in the two distributions represented by	(05)	L4	CO1

Academic year 2024-2025 (Even Sem)

	the following boxplots. Assume the two boxplots have the same scale. 			
b	In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1,2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins what is the probability that it is defective and find the conditional probability that it was produced by machine 3	(05)	L4	CO1
3	Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products.			
	a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review?	(04)	L3	CO2
	b) If a product attains a good review, what is the probability that it will be a highly successful product?	(03)	L3	CO2
	c) If a product does not attain a good review, what is the probability that it will be a highly successful product?	(03)	L3	CO2
4	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G)	(10)	L3	CO2
5	Suppose the excess length in a manufactured part in mm is recorded as below.	(10)	L2	CO1

19	22	6	29	57	14	12	24	6	37
19	19	18	12	11	17	15	17	13	56

i. Compute the mean, median and mode for the data set.
ii. Draw a histogram with 4 classes and a box plot, and interpret.

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	7	3	-	-	1	2	7	-	-	-
	Test		30	20	-	-	-	10	20	30	-	-

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Academic year 2024-2025 (Even Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

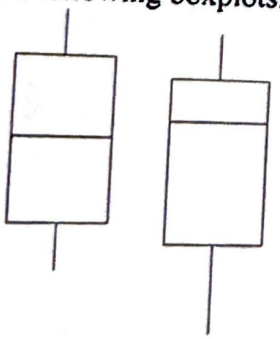
Date	**01/04/2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM241AT**	**Duration**	**30 + 90 Min**
Sem	**IV**	**CIE – I**	
STATISTICS FOR DATA ANALYTICS (scheme and solution)			

Note:

1. **Answer all the Questions.**

Sl. No.	Questions	M	BT	CO
	Part – A			
1	40th percentile in a given set of 140 data values occurs at position ___56th_____ 40th percentile in 140 data values • Position = (40/100) × 140 = 56th • Answer: 56th position	(1)	L2	CO1
2	<table><tr><td>Class</td><td>21"-30"</td><td>31"-40"</td><td>41"-50"</td><td>51"-60"</td><td>61"-70"</td></tr><tr><td>No of occurrences</td><td>12</td><td>16</td><td>22</td><td>25</td><td>18</td></tr></table>The median in the above data lies in the class __41 -50'' class_____ Cumulative frequencies: 12, 28, 50, 75, 93 → Median is at 47th position Answer: 41"-50"	(1)	L2	CO1
3	According to Bayes' rule, P(A|B) = ___, for two events in a sample space S. P(A|B)=P(B)*P(B|A) / P(A)	(1)	L1	CO1
4	X = the no. coming up when a 6-faced die is thrown. Y = no. of heads when a coin is tossed. The possible values of X - Y are _0 to 6_____.	(1)	L3	CO2
5	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A U B U C U D, and each element of S is equally likely to happen, then P(A′∩ B) =__ S=A∪B∪C∪D={1,2,3,4,5,6,7,8,9}, total 9 A′∩B={6,8}A' ∩ B = \{6,8\}A′∩B={6,8} → P = 2/9.	(1)	L3	CO2
6	P(X <= 3| Y= 0) is ___ , for X and Y as in Question no. 1.4 When Y=0 → Coin = T, X=1 to 6 X ≤ 3: 1,2,3 → 3 favourable out of 6 **Ans: 36=0.5\frac{3}{6} = 0.563=0.5**	(1)	L3	CO2
7	A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective? _____ Without replacement: (25/100)×(24/99)=(600/9900) ≈0.0606	(2)	L3	CO1
8	Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: <table><tr><td rowspan="2">Type of unsaturation</td><td colspan="2">Type of oil</td></tr><tr><td>Canola</td><td>Corn</td></tr><tr><td>Mono</td><td>7</td><td>13</td></tr><tr><td>Poly</td><td>93</td><td>77</td></tr></table>If a bottle of oil is selected at random, what is the probability that it belongs to	(2)	L3	CO1

Academic year 2024-2025 (Even Sem)

	the polyunsaturated category? Total =190 Prob =170/190 =17/19 = 0.8947			
1	Part –B Fourteen measurements of furnace temperatures (in ^{0}F) are recorded on successive batches in a semiconductor manufacturing process as follows: 953, 950, 948, 955, 951, 949, 957, 954, 955, 949, 952, 931, 953, 955. a) Calculate sample mean, sample variance, and standard deviation. b) Find the mode and median. How much could the largest temperature measurement be increased without changing the median value? c) Construct a box plot of the data, showing clearly the outlier bounds. d) Comment on the symmetry of the data. Mean = (sum of 14 values)/14 = 13342 / 14 = **952.57°F** Median = 952 Mode = 955 Sample variance s2s^2s2 = approx. **43.45** Standard deviation ≈ **6.59** Largest temperature that doesn't change median = 955 (since 7th and 8th values are 953, 955) Boxplot: • Outliers = values outside Q1–1.5IQR or Q3+1.5IQR • 931 is an outlier Symmetry: Slight left skew due to low outlier (931)	(10)	L3	CO1
2 a	Describe any similarities or differences in the two distributions represented by the following boxplots. Assume the two boxplots have the same scale.  Look at median (central tendency), IQR (spread), symmetry (skewness) Eg: Boxplot A has longer right whisker → positive skew	(05)	L4	CO1
b	In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1,2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins what is the probability that it is defective and find the conditional probability that it was produced by machine 3 P(D)=0.35×0.02+0.25×0.01+0.40×0.03=0.007+0.0025+0.012=0.0215 P(M3|D) =0.40×0.030.0215≈0.558P	(05)	L4	CO1

3	Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products. a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review? b) If a product attains a good review, what is the probability that it will be a highly successful product? c) If a product does not attain a good review, what is the probability that it will be a highly successful product? P(Good Review) = 0.4×0.95 + 0.35×0.6 + 0.25×0.1 = 0.38 + 0.21 + 0.025 = **0.615** P(Highly Successful | Good) = 0.38 / 0.615 ≈ **0.617** P(Highly Successful | Not Good) = 0.02 / 0.385 ≈ **0.052**	(04) (03) (03)	L3 L3 L3	CO2 CO2 CO2
4	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G) Total = 10 P(D|C) = 3/4 = 0.75 P(D|G) = 2/6 = 1/3 P(G|D) = 2/5 = 0.4 P(C|S) = 1/5 = 0.2 P(G) = 6/10 = 0.6	(10)	L3	CO2
5	Suppose the excess length in a manufactured part in mm is recorded as below. 19 22 6 29 57 14 12 24 6 37 19 19 18 12 11 17 15 17 13 56 i. Compute the mean, median and mode for the data set. ii. Draw a histogram with 4 classes and a box plot, and interpret. • Mean ≈ 22.15 mm • Median = 17 • Mode = 19 • Histogram: 4 classes → 0–15, 16–30, 31–45, 46–60 • Box plot: Outliers likely at 56, 57 • Interpretation: Positive skew due to high values	(10)	L2	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max Marks	7	3	-	-	1	2	7	-	-	-
	Test		30	20	-	-	-	10	20	30	-	-

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Academic year 2023-2024 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18.06.2024**	**Maximum Marks**	**50**
Course Code	**IM241AT**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.

<table>
<tr><th>Sl. No.</th><th>Questions</th><th>M</th><th>BT</th><th>CO</th></tr>
<tr><td>1</td><td>Calculate Mean, Mode and Median for following group data
<table>
<tr><th>Class Interval</th><th>Frequency</th></tr>
<tr><td>1 under 3</td><td>4</td></tr>
<tr><td>3 under 5</td><td>12</td></tr>
<tr><td>5 under 7</td><td>13</td></tr>
<tr><td>7 under 9</td><td>19</td></tr>
<tr><td>9 under 11</td><td>7</td></tr>
<tr><td>11 under 13</td><td>5</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>2</td><td>Construct a box plot for following data:
<table>
<tr><td>87</td><td>85</td><td>84</td><td>84</td><td>82</td><td>82</td><td>82</td><td>81</td><td>81</td><td>81</td></tr>
<tr><td>80</td><td>79</td><td>79</td><td>77</td><td>76</td><td>75</td><td>74</td><td>74</td><td>74</td><td>73</td></tr>
<tr><td>73</td><td>73</td><td>73</td><td>72</td><td>72</td><td>71</td><td>71</td><td>71</td><td>70</td><td>69</td></tr>
<tr><td>69</td><td>68</td><td>68</td><td>65</td><td>65</td><td>64</td><td>64</td><td>63</td><td>62</td><td>62</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>3</td><td>Assume the following are the marks obtained in Statistics for Data Analytics CIE1. Calculate Mean, Median, Mode, Variance and Standard Deviation for the data.
<table>
<tr><td>33</td><td>43</td><td>44</td><td>45</td><td>46</td><td>49</td><td>31</td><td>32</td><td>34</td><td>36</td></tr>
<tr><td>39</td><td>26</td><td>29</td><td>22</td><td>22</td><td>23</td><td>25</td><td>11</td><td>19</td><td>18</td></tr>
<tr><td>19</td><td>15</td><td>23</td><td>45</td><td>44</td><td>29</td><td>39</td><td>38</td><td>28</td><td>27</td></tr>
<tr><td>26</td><td>28</td><td>35</td><td>39</td><td>38</td><td>38</td><td>37</td><td>37</td><td>28</td><td>29</td></tr>
</table></td><td>10</td><td>L3</td><td>CO1</td></tr>
<tr><td>4</td><td>Construct a Histogram for the following data.
35, 339, 650, 864, 1025, 1261, 38, 340, 655, 883, 1028, 1280, 48, 395, 669, 883 1036, 1290, 53, 457, 703, 890, 1044, 1312, 70, 478, 730, 934, 1087, 1341, 99 501, 763, 951, 1091, 1355, 138, 521, 788, 969, 1126, 1357, 164, 556, 789, 985 1176, 1360, 220, 583, 789, 993, 1199, 1414, 265, 595, 802, 997, 1199, 1436, 272, 596, 822, 999, 1237, 1479, 312, 604, 851, 1018, 1242, 1492</td><td>10</td><td>L3</td><td>CO2</td></tr>
<tr><td>5</td><td>Write a Note on Following
i. Population and Sample
ii. Parameter and Statistics
iii. Descriptive and Inferential Statistics
iv. Probability and Probability distribution</td><td>10</td><td>L2</td><td>CO2</td></tr>
</table>

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	30	20	--	--	--	10	40	--	--	--

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DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	5th September 2023	Maximum Marks	50
Course Code	21IM41	Duration	90 Min
Sem	IV	Improvement Test	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.
2. Use of statistical tables permitted.

Q. No.	Questions	M	BT	CO
1.	An engineer who is studying the tensile strength of a steel alloy intended for use in golf club shafts knows that tensile strength is approximately normally distributed. A random sample of 12 specimens has a mean tensile strength of 3250 psi. i. Test the hypothesis that mean strength is 3500 psi. Use 1 % significance level. Standard deviation is known to be 60 psi ii. What is the smallest level of significance at which you would be willing to reject the null hypothesis in part (i)? iii. Explain how you could answer the question in part (i) with a two-sided confidence interval on mean tensile strength. iv. How will the test be performed if the standard deviation is not known?	(10)	L3	CO3
2.	Inter arrival times in minutes between successive customers in a hotel is exponentially distributed with a mean of 2 minutes. i. Compute the probability that inter arrival time between two customers is at most 2 minutes. ii. What is the probability that at least 2 customers arrive in one minute? iii. What is the mean-time till every 3rd customer arrives? iv. What is the probability of no customers arriving in a span of 3 minutes?	(10)	L3	CO3
3.	i. What are the hypothesis statements and the statistic to be used in a two-sided test of variance of a population? What are the assumptions to use this statistic and what is the rejection region for this test? ii. State the probability of a t-random variable with 25 degrees of freedom taking on any value between 2.06 and 3.45 iii. What is the value x such that a chi-square random variable with 15 degrees of freedom takes on at the most a value of x with 95% likelihood?	(10)	L2	CO4
4.	For the sample (5,2,3,1,0,0,1,3,3) from a population, i. Construct a 90 % confidence interval on population mean if the population standard deviation is known to be 2. ii. What should the sample size be in order to keep the width of the confidence interval to less than 2? iii. Construct a 95 % confidence interval on population mean if the population variance is unknown.	(10)	L3	CO4
5.	An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find i. the probability that a bulb burns between 778 and 834 hours. ii. Probability that a bulb has a life of at least 1000 hours iii. What percent of bulbs have their life within ± 2σ from their own mean life, where σ is the standard deviation?	(10)	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Test Max Marks	--	--	20	30	--	10	40	--	--	--

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CIE - I - 2023

Academic year 2022-2023 (Even Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**5th July 2023**	**Maximum Marks**	**50**
Course Code	**21IM41**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE-I**	

STATISTICS FOR DATA ANALYTICS

Note:
1. Answer all the Questions.

Q. No.	Questions	M	BT	CO
1.	20 weight measurements in a process were recorded as: 61 24 23 27 29 59 47 38 49 12 36 75 64 83 92 18 29 18 20 19 i. Draw a box plot for the data. ii. What are the range and the Inter Quartile Range for the data? iii. Are there any outliers? If yes, what are they? iv. Draw the stem and leaf diagram and comment on the distribution of data	(10)	L3	CO1
2.	Three machines A, B and C produce respectively 50%, 30% and 20% of total number of items in a factory. The proportion of defective output of these machines are 0.03, 0.04 and 0.05 respectively. If an item is selected at random find (i) The probability that the item selected is defective. (ii) If through an action of improvement, the percentage of defective output is reduced to 2%, 2% and 3% respectively. Revise the probability estimate of (i) above.	(10)	L3	CO1
3.	A multiple-choice test contains 24 questions, each with four answers. Assume a student just guesses on each question. (a) What is the probability that the student answers at least 21 questions correctly? (b) What is the probability the student answers at most 10 questions correctly?	(05) (05)	L3	CO2
4.	Assume that a box contains 10 balls distributed as follows: Three are colored(C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S) a) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G)	(10)	L3	CO2
5.	A random variables X is distributed as $f(x) = x/k$, $1 \le x \le 4$ a) What is the probability that X will take on a value of at most 2? b) Plot the probability density function of X. c) Derive the cumulative distribution of X and plot the same. d) What is the median of X? Is it the same as mean of X?	(03) (02) (03) (02)	L3	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	30	--	--	--	--	50	--	--	--

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